Inequalities — problem 7.2

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Prove that, for $a, b, c > 0$,

(a) $\dfrac{a^2+b^2}{a+b} \geq \dfrac{a+b}{2}$, (b) $\dfrac{a^3+b^3+c^3}{a^2+b^2+c^2} \geq \dfrac{a+b+c}{3}$, (c) $\dfrac{a+b+c}{3} \geq \sqrt{\dfrac{ab+bc+ca}{3}} \geq \sqrt[3]{abc}$.